Unit 5 Pre-Test

Statistics: Center, Spread & Regression

Standards assessed: NC.M1.S-ID.1 · S-ID.2 · S-ID.3 · S-ID.6 · S-ID.7 · S-ID.8 · S-ID.9

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.M1.S-ID.1 · NC.M1.S-ID.2 · NC.M1.S-ID.6 Read data displays and find center and spread.

• *I can read dot plots, histograms, box plots, and scatter plots, and find the mean, median, and range.*

Answer each question. Use the picture to help you.

How to type your answers

number `6.5` negative `-2` word `median` letter `B`

1. *Mean*

Find the mean of 4, 7, 9, 10, 15.



mean = _____

answer: 9

2. *Median*

Find the median of 3, 8, 5, 12, 9, 6.



median = _____

answer: 7

3. *Range*

Find the range of 12, 18, 7, 25, 14.

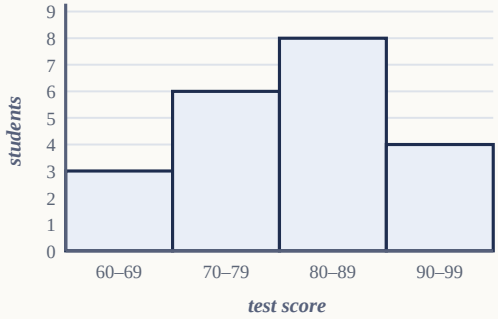


range =

answer: 18

4. *Histogram*

The histogram shows test scores. How many students are in the data set?



students =

answer: 21 students

5. *Box plot*

Use the box plot. What is Q3 (the upper quartile)?

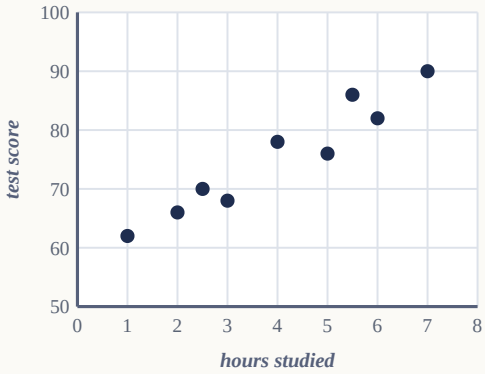


Q3 =

answer: 11

6. *Association*

Describe the association: positive, negative, or none?



Answer:

answer: positive

SCORE **4.0** **Target Content**

NC.M1.S-ID.2 · NC.M1.S-ID.3 · NC.M1.S-ID.6 · NC.M1.S-ID.8 Use IQR and outliers, predict with a trend line, and read residuals and r.

• *I can find the IQR, test for outliers, predict with a trend line, find a residual, and match r to a scatter plot.*

Solve each problem. Show your work on paper.

7. *IQR*

Find the interquartile range (IQR) of 2, 4, 5, 7, 8, 10, 12, 15.

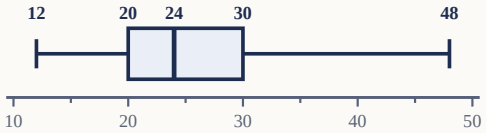


IQR = _____

answer: 6.5

8. *Outlier?*

Use the 1.5 × IQR rule. Is 48 an outlier? Write yes or no.



Answer: _____

answer: yes (upper fence = 30 + 15 = 45)

9. *Best center*

Data: 10, 13, 16, 19, 52. Which describes the center better: the mean or the median?

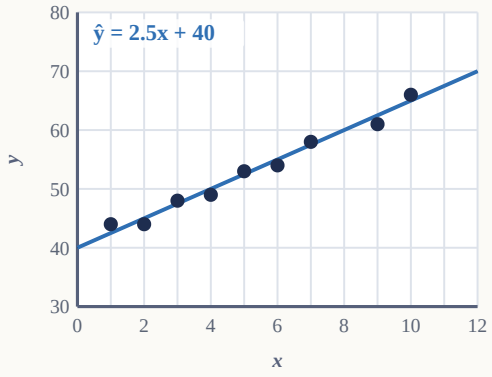


Answer: _____

answer: the median (52 is an outlier)

10. *Predict*

The trend line is ŷ = 2.5x + 40. Use the equation to predict y when x = 11.

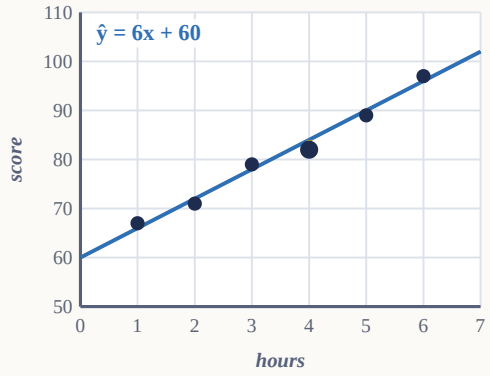


ŷ = _____

answer: 67.5

11. *Residual*

The trend line is $\hat{y} = 6x + 60$. A student studied 4 hours and scored 82. Find the residual.

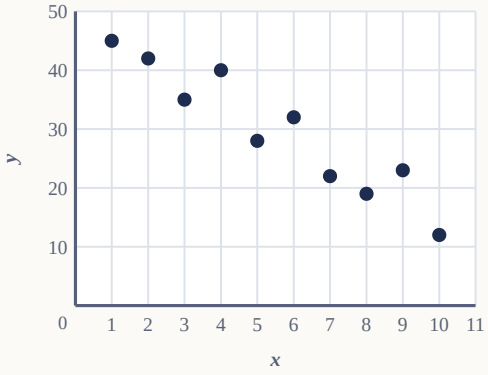


residual = _____

answer: −2

12. *Correlation*

Which value of r fits this scatter plot best?
−0.95, −0.3, 0.3, 0.95



r = _____

answer: −0.95

SCORE **5.0** **Advanced Content**

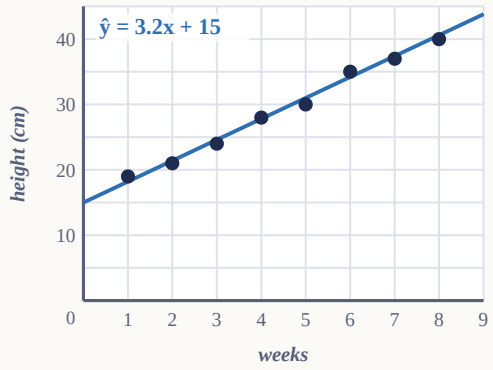
NC.M1.S-ID.2 · NC.M1.S-ID.7 · NC.M1.S-ID.8 · NC.M1.S-ID.9 Interpret models and compare data sets.

• *I can use slope and intercepts in context, compare spread, compare correlation strength, and tell correlation from causation.*

Solve each problem. Some answers are words.

13. *Interpret slope*

ŷ = 3.2x + 15 predicts plant height (cm) after x weeks. Use the slope: how many cm does the model say the plant grows from week 2 to week 6?

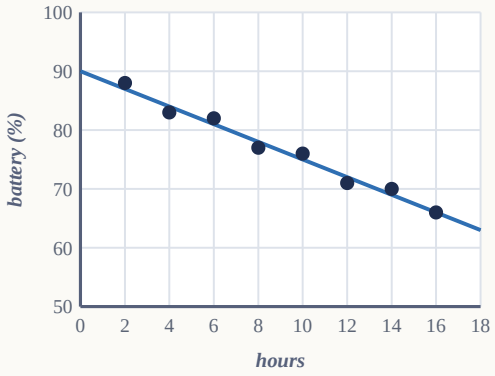


cm = _____

answer: 12.8 cm

14. *Interpret intercepts*

ŷ = −1.5x + 90 predicts phone battery (%) after x hours. The 90 means the battery starts at 90%. Use the model: after how many hours is the battery at 0%?

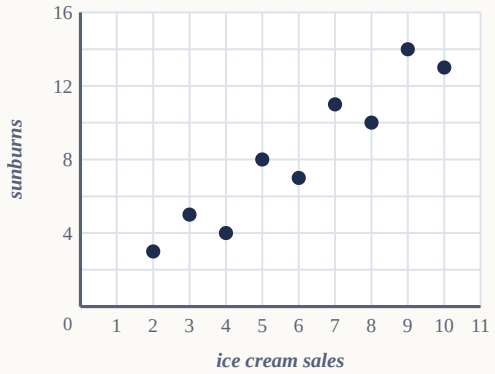


hours = _____

answer: 60 hours

15. *Causation?*

Ice cream sales and sunburns have a strong positive correlation. Does eating ice cream **cause** sunburns? Write yes or no.

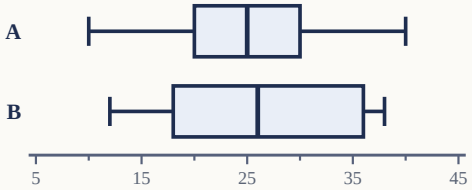


Answer:

answer: no (hot, sunny weather drives both)

16. *Compare spread*

Which class has the greater spread, using the IQR: A or B?

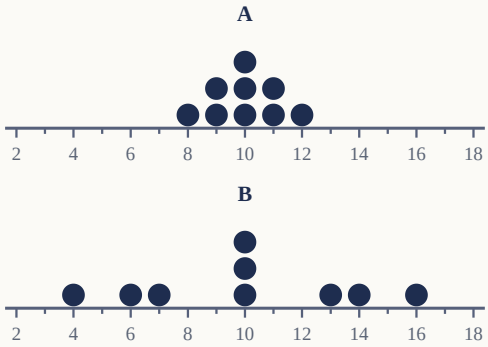


Answer:

answer: B (IQR 18 vs. 10)

17. *Standard deviation*

Both data sets have a mean of 10. Which has the greater standard deviation: A or B?

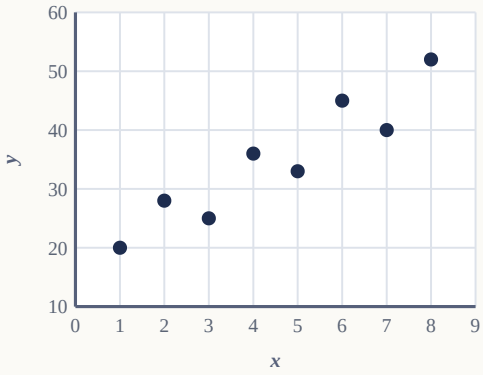


Answer:

answer: B (the data are farther from the mean)

18. *Correlation strength*

Set A is graped. Its r = 0.93. Set B has r = −0.97. Which set has the **stronger** linear association: A or B?



Answer:

answer: B (|−0.97| is closer to 1)